



MOBNET in passive seismic experiments Part II – quality control

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Quality control and State of Health



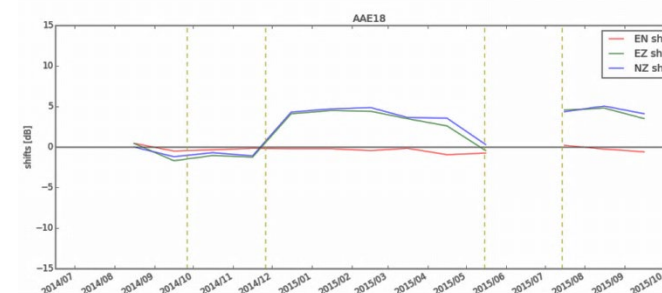
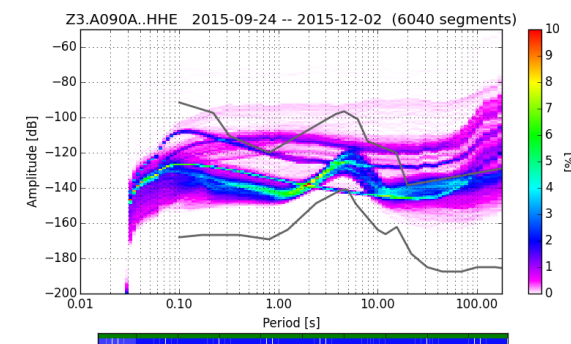
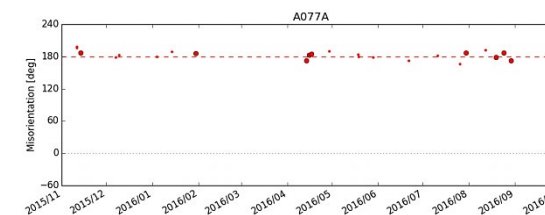
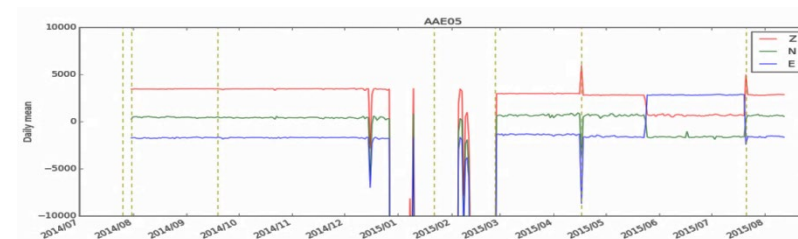
hardware methods



Data availability
Metadata
Signal clipping
Drift of sensor mass position
Amplitude gain imperfection
Noise in signals
Glitches in signal
Sensor orientation
Reversed or interchanged components
Timing issues

...

software methods



Station metadata - station siting, instrumentation

• network code	Y8	}	must be same as in mseed	• sensor orientation
• station code	AVR			- azimuth
• location code	00			- dip
• channel codes	HHZ, HHN, HHE			• responses
• sample rating	100 [Hz]			- datalogger
• latitude	43.11949 [deg]			- sensor
• longitude	27.66911 [deg]			
• elevation	292 [m]			... and more
• time epochs	startDate, endDate			

Metadata file formats

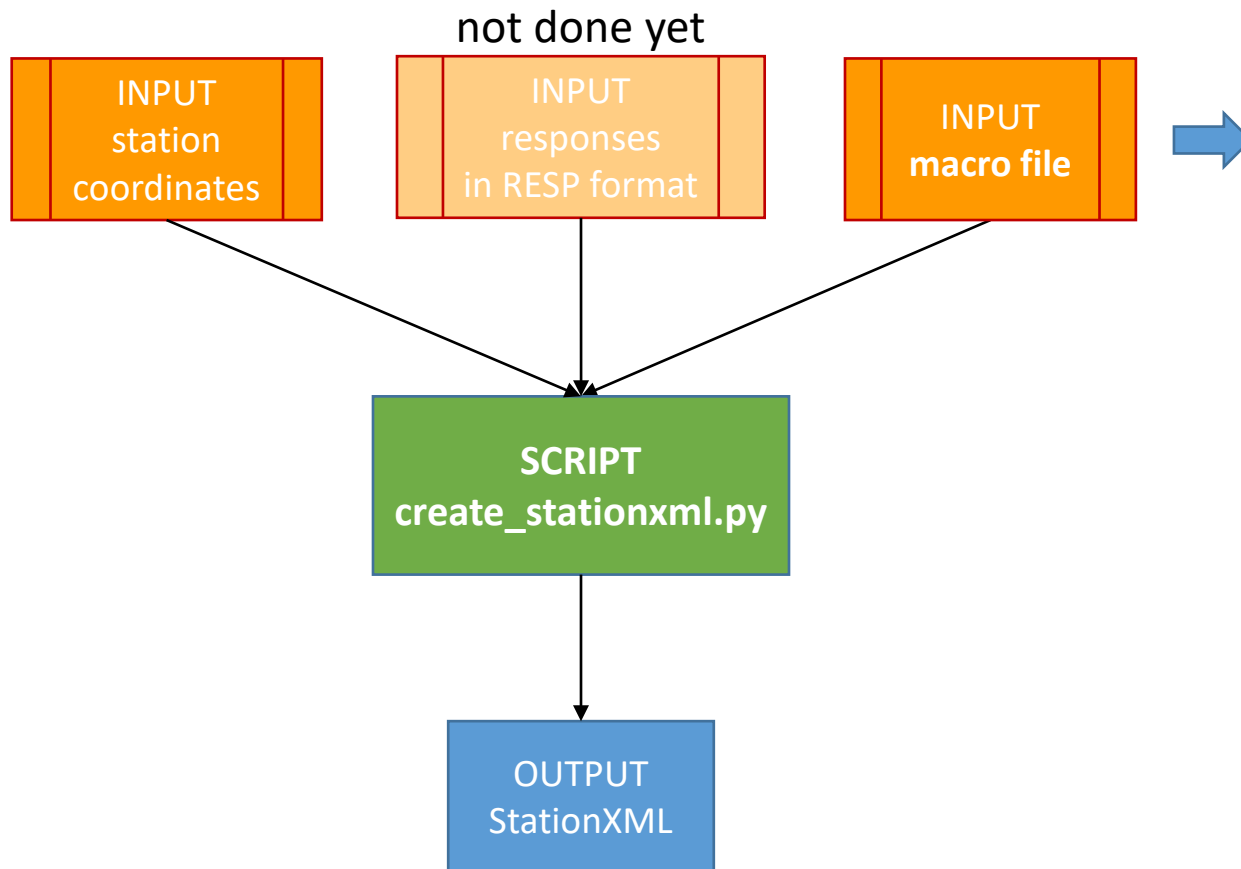
- Dataless SEED (binary)
https://ds.iris.edu/ds/nodes/dmc/data/formats/#dataless_seed
- RESP format (ASCII)
 - ASCII representation of Dataless SEED with only instrument response information<https://ds.iris.edu/ds/nodes/dmc/data/formats/resp/>
- **StationXML** (ASCII)
 - a modern alternative to Dataless SEED<http://www.fdsn.org/xml/station/>

All standards promoted by the International Federation of Digital Seismograph Networks (FDSN;
<https://www.fdsn.org/>)

Creating station metadata

- PDCC (IRIS, last version 3.8.1, 2015-09-24)
<https://ds.iris.edu/ds/nodes/dmc/software/downloads/pdcc/>
 - window editor, input & output: Dataless SEED, RESP, manual/click input
 - bug: unsorted response stages in output, [stationxml-converter.jar](#) ver. 1.0.9 (IRIS) cannot convert unsorted Dataless SEEDs correctly -> **not recommended!**
- Creating a StationXML file from Scratch (ObsPy 1.1+)
https://docs.obspy.org/tutorial/code_snippets/stationxml_file_from_scratch.html
- Script `create_stationxml.py` (by Luděk Vecsey, ObsPy 1.0+)

Script create_stationxml.py



Input macro file

```
<network> XT

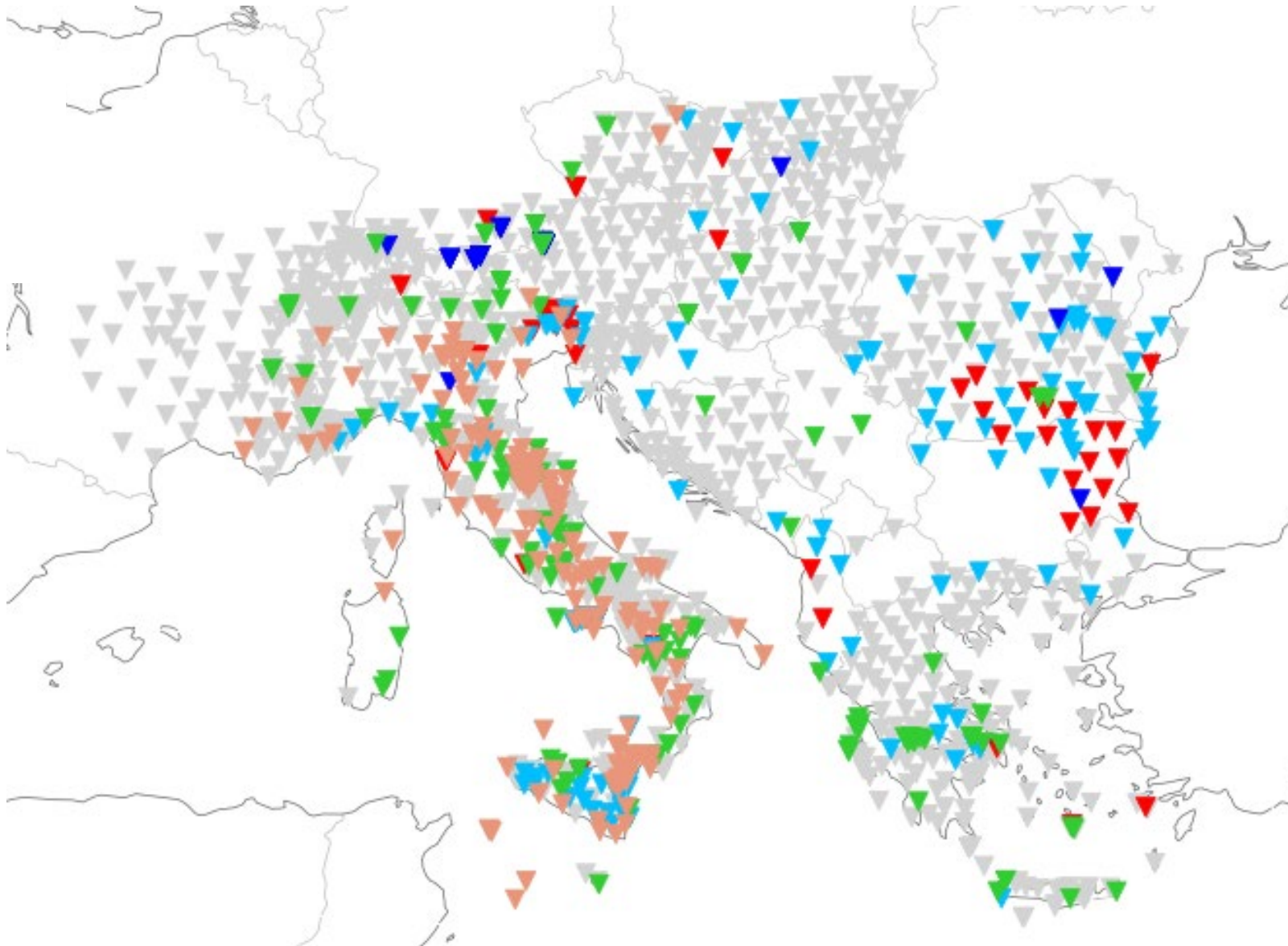
<station> AAE01 <channel> HH <sample_rate> 100.
          <seismometer> STS-2/II
          <start> 2014-07-23T11:09:00
          <receiver> GAIA2T
          <start> 2014-11-21T12:00:00
          <receiver> Q330S
          <end> 2015-10-22T08:43:00

<station> AAE02 <channel> HH <sample_rate> 100.
          <seismometer> STS-2/III
          <receiver> GAIA1/65Hz
          <start> 2014-06-24T10:25:00 <#orientation> 7.4
          <start> 2014-10-21T08:10:00 <#orientation> 0.4
          <end> 2015-09-03T07:42:00

<station> AAE03 <channel> HH <sample_rate> 100.
          <seismometer> CMG-40T
          <receiver> GAIA1/25Hz
          <#seism_gain:N> 794. <#seism_gain:E> 804.
          <start> 2014-06-24T12:51:00 <#orientation> 4.0
          <start> 2014-10-21T07:13:40 <#orientation> 0.2
          <end> 2015-09-03T06:44:00
```

Check of _ADRIAARRAY metadata

Metadata downloaded
from EIDA 29.-30.3.2023



▲ metadata warnings
(15.3 %)

▲ errors in metadata
(3.6 %)

▲ wrong band code
(8,4 %)

▲ different corner periods
in metadata and the list
(8.8 %)

▲ switched broad-band
and short-period
(1.0 %)

total: 37 %

Daily means – fast and powerful tool for QC

Probabilistic power spectral density (PPSD)

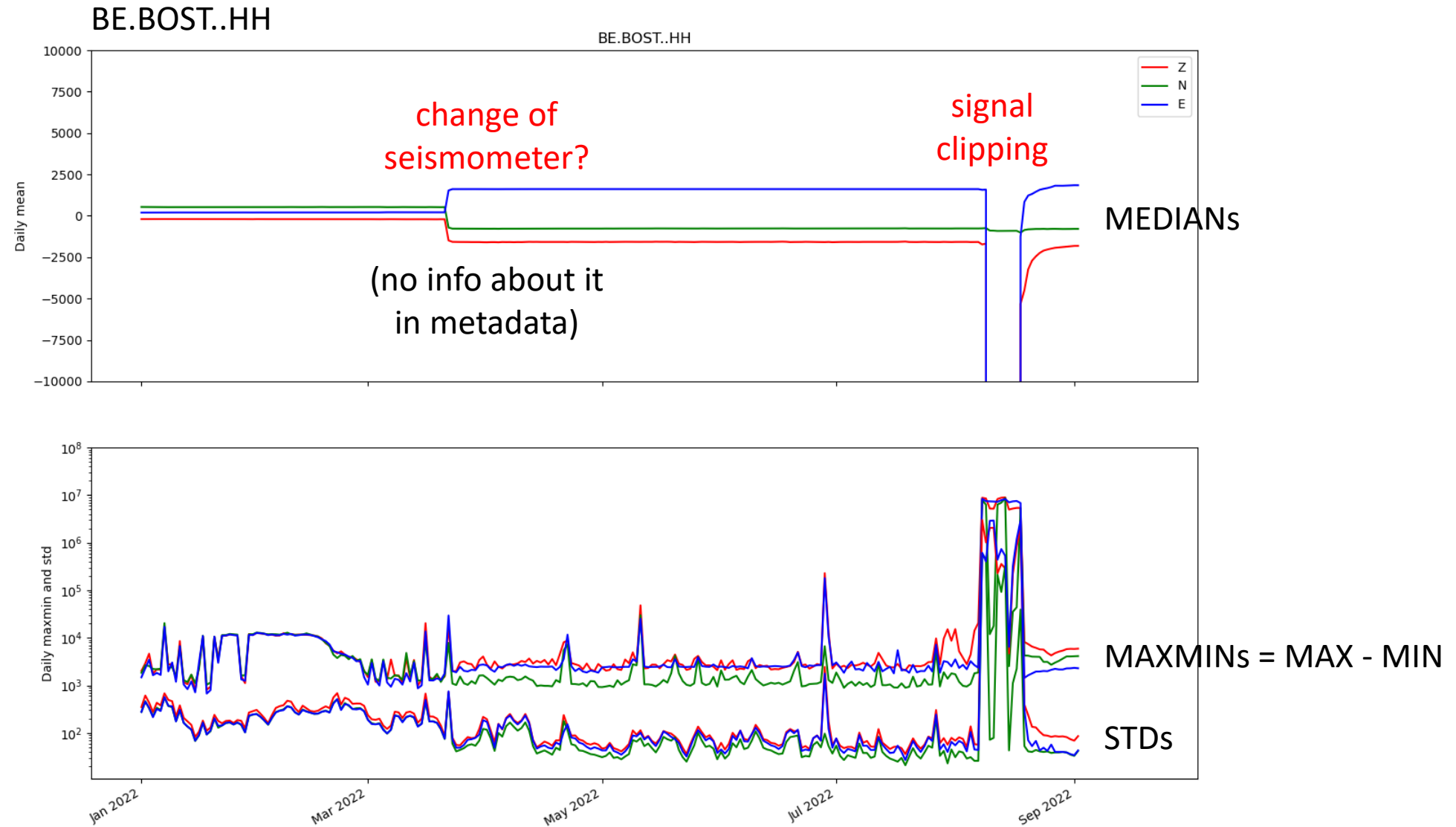
- time-frequency representation of the signal
- not included in EIDA WFCatalog (yet)
- need to download continuous data
- long computation of PPSD
- depends on metadata (changes/updates!)

Metrics in WFCatalog

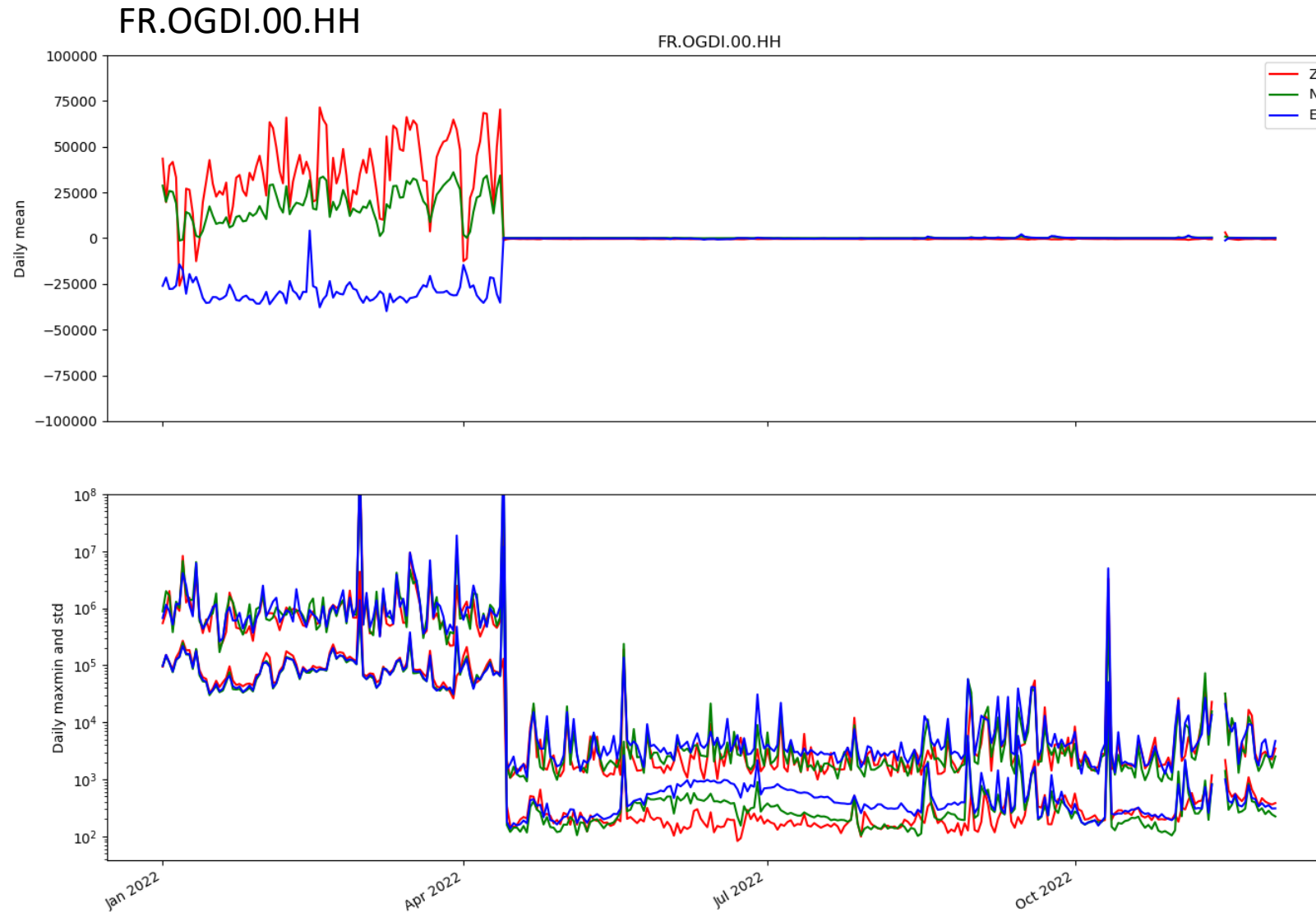
- no frequency analysis
- easy and fast to download from EIDA
- doesn't depend on metadata
- metrics:
 - daily median
 - daily std
 - daily max, min

...

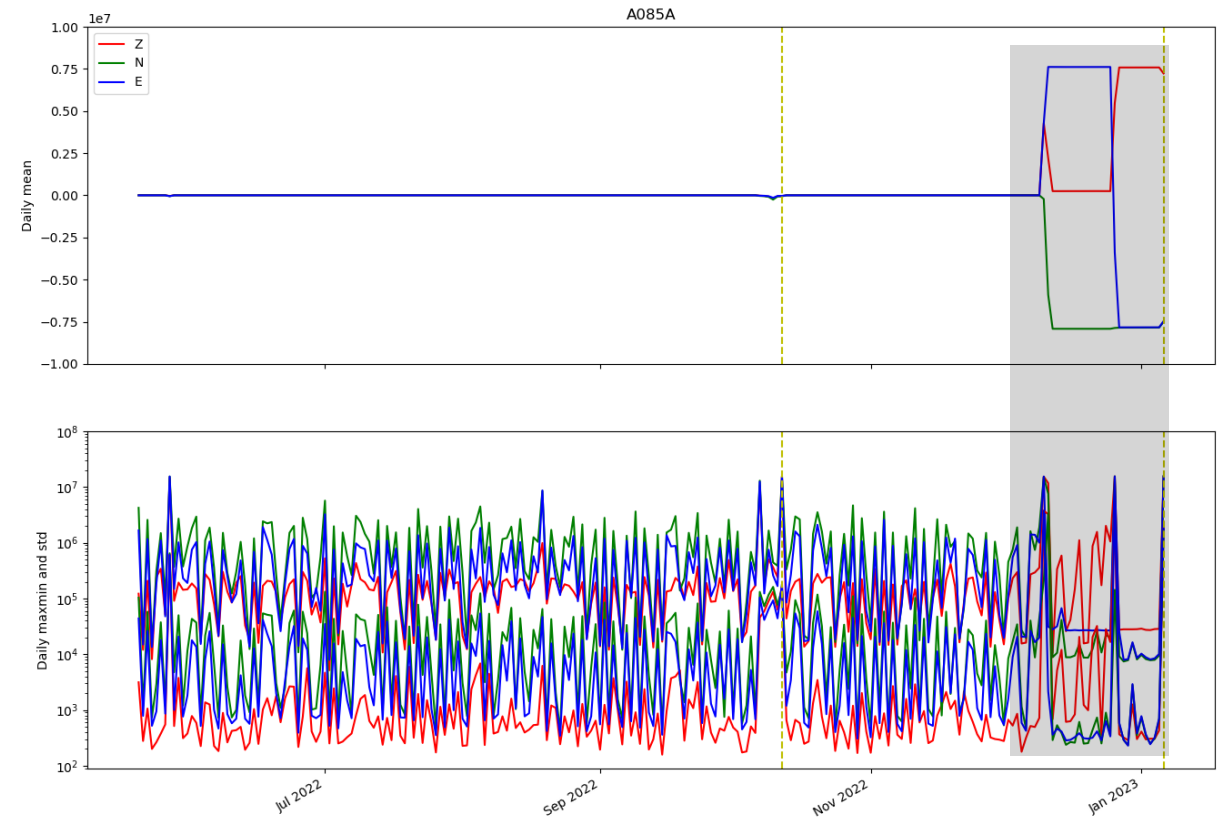
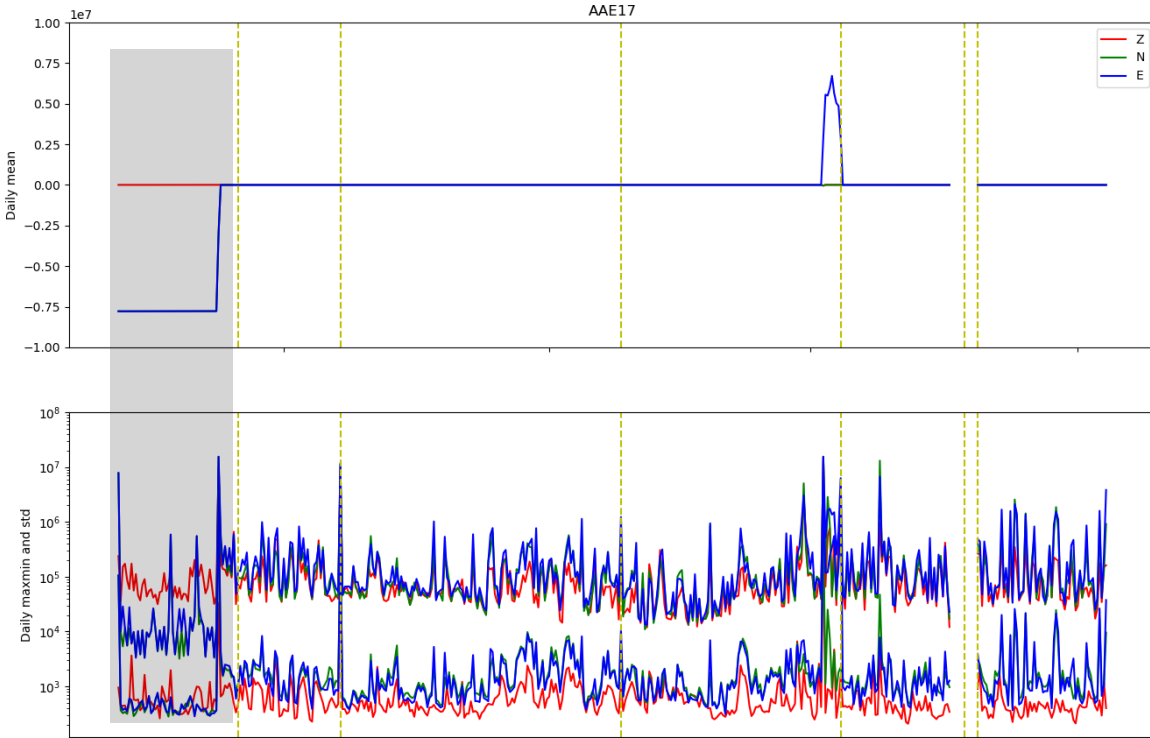
Change of seismometer, signal clipping



Improvement of station registration



Saturated signal offset, signal clipping



What is signal maximum?

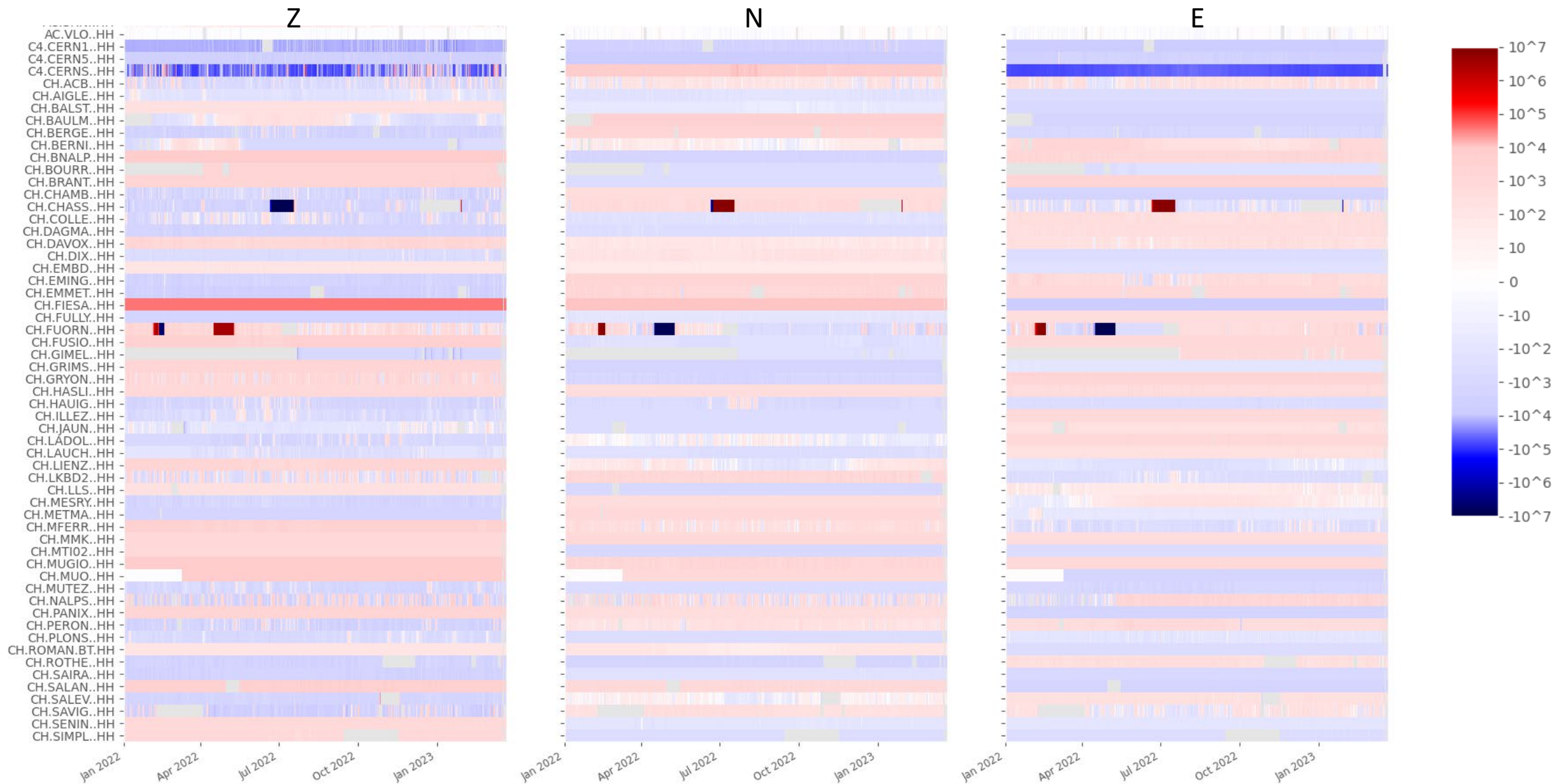
24 bit ADC = 8.4×10^6

32 bit ADC = 2×10^9

Gray areas:

saturated offsets $\rightarrow$ saturated amplitudes

Saturation of signal daily medians/offsets



Saturated medians - map

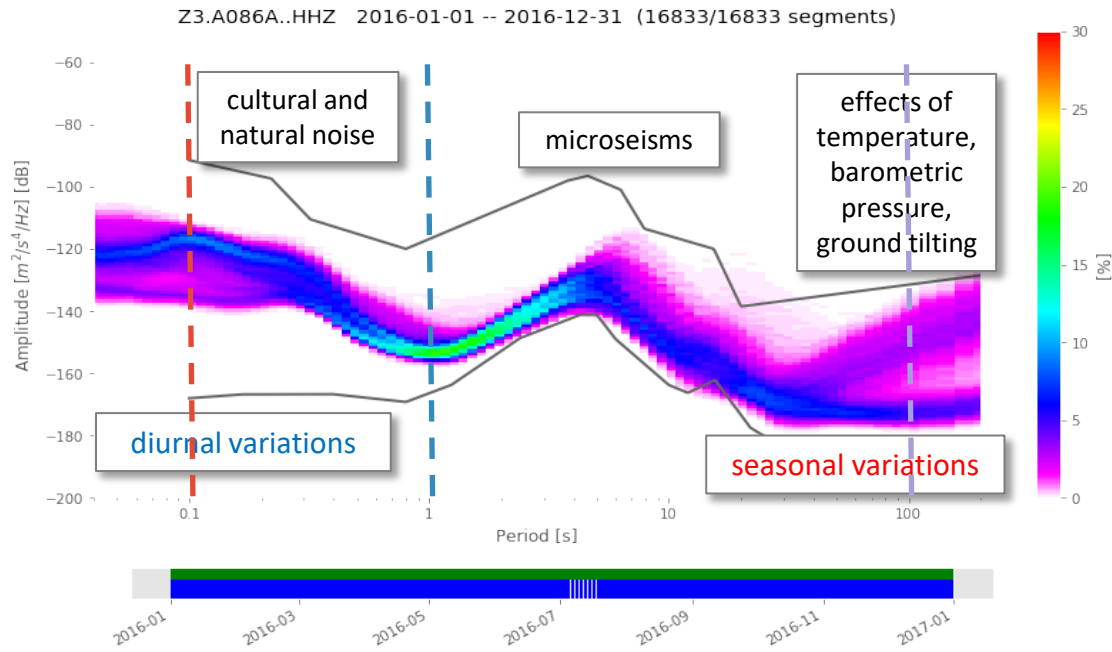
Tested period:
2022-01-01 – 2023-03-21
445 days

Saturation condition:
 $\text{abs}(\text{median}) > 1\text{e}6$

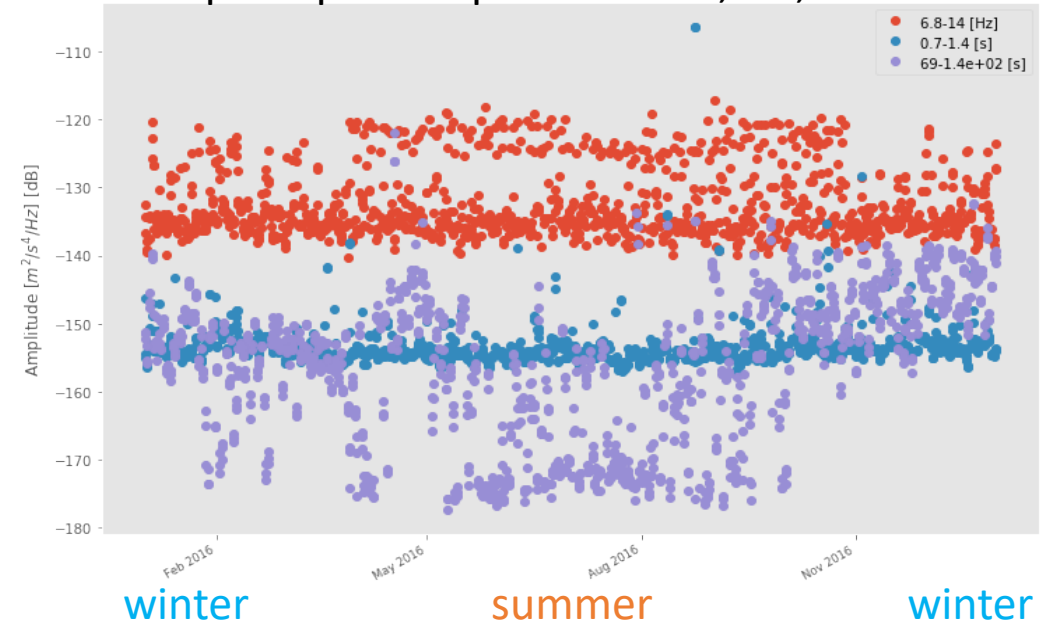


- ▲ all stations (861)
- ▲ 10+ days: 79/861 9 %
- ▲ 100+ days: 43/861 5 %
- ▲ 300+ days: 28/861 3 %

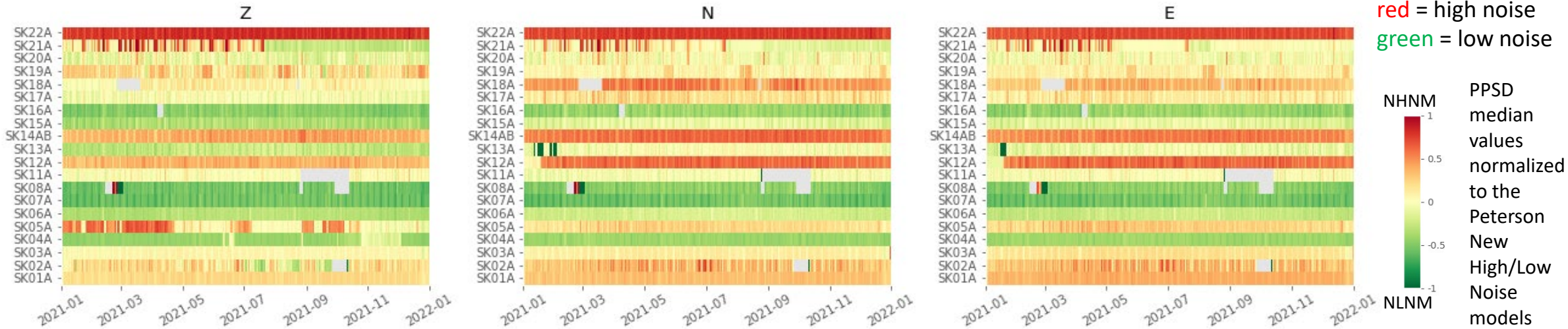
Probabilistic power spectral density (PPSD)



Temporal plot for periods 0.1 s, 1 s, 100 s



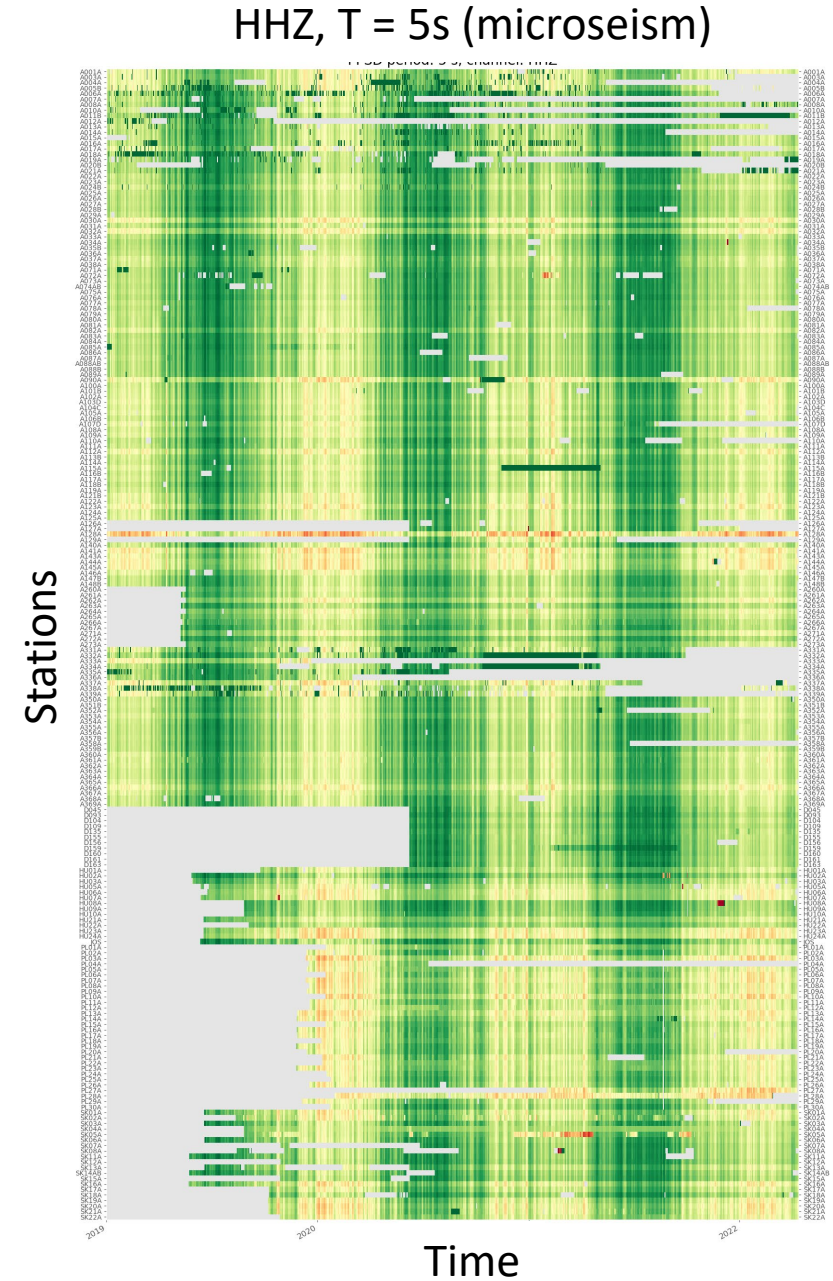
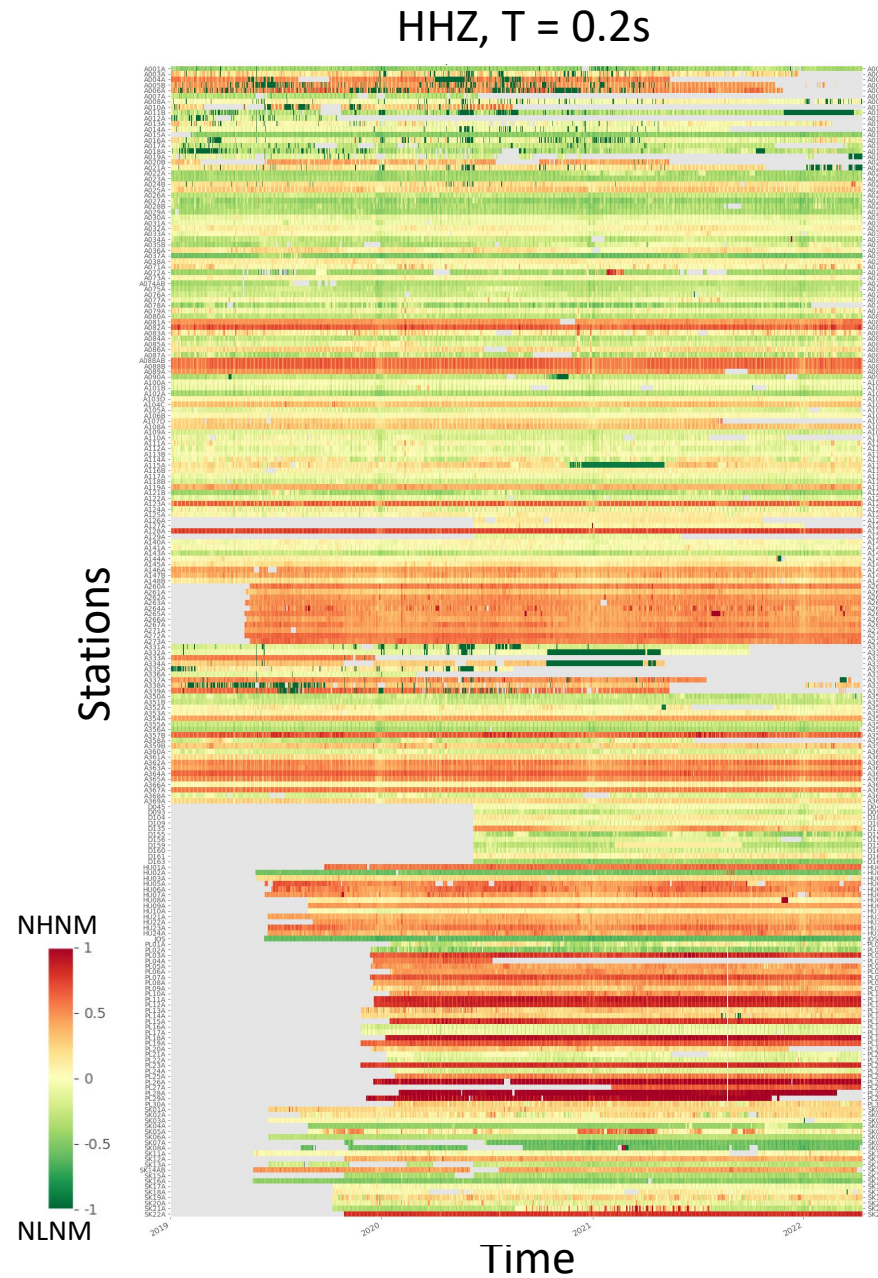
PPSDs
on
period
0.2 s



PPSDs for the PACASE experiment 2019-2022

- thank Antje Schlömer for PPSD computation

Schlömer et al., 2023, submitted

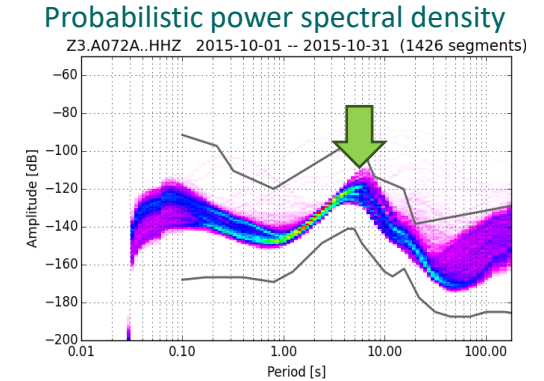
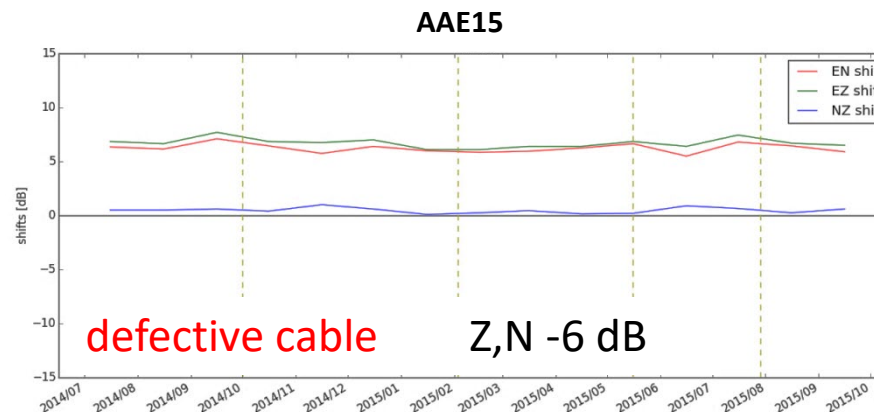
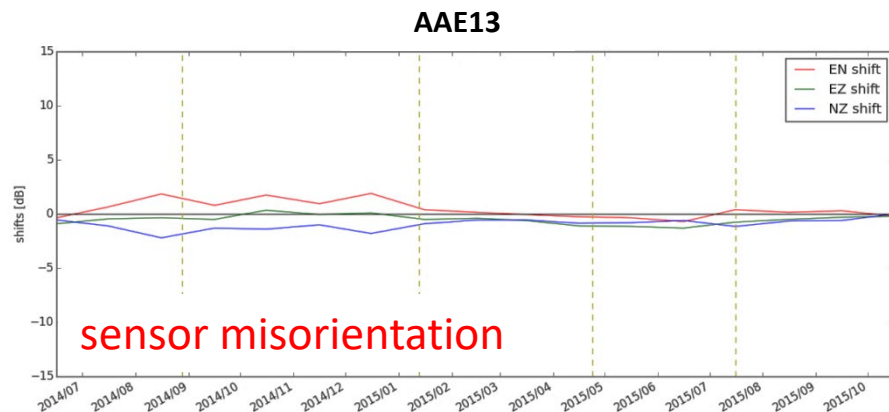
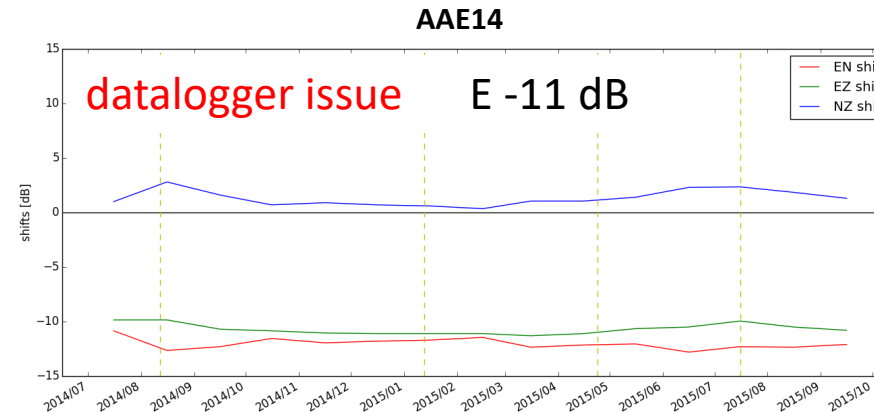
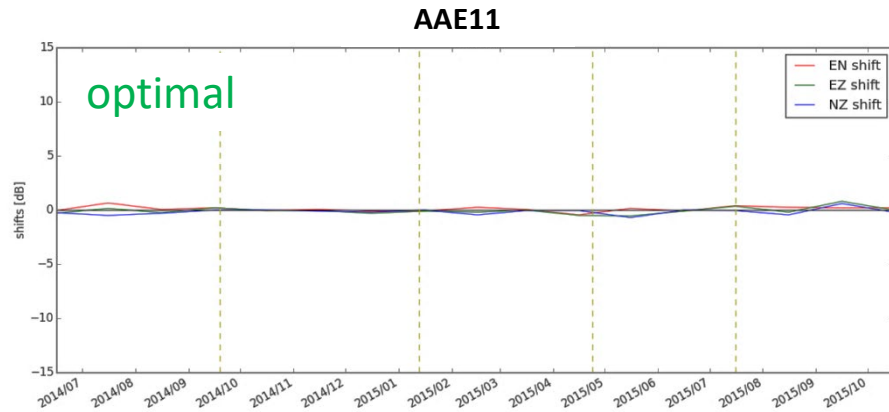


Detection of amplitude gain issues by the PPSD method

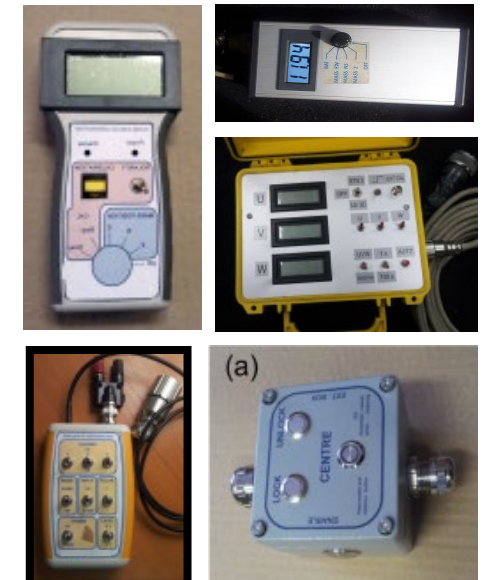
It compares ratios of PPSDs between the three components in a range of 4-8 seconds (where the secondary microseisms are substantially larger than noise from local sources).

A global and local trends of the ratios are removed.

Double check - stations with amplitude misgains are tested by special control devices for seismometers (CMG, STS) and GAIA DASS (developed in IG CAS).

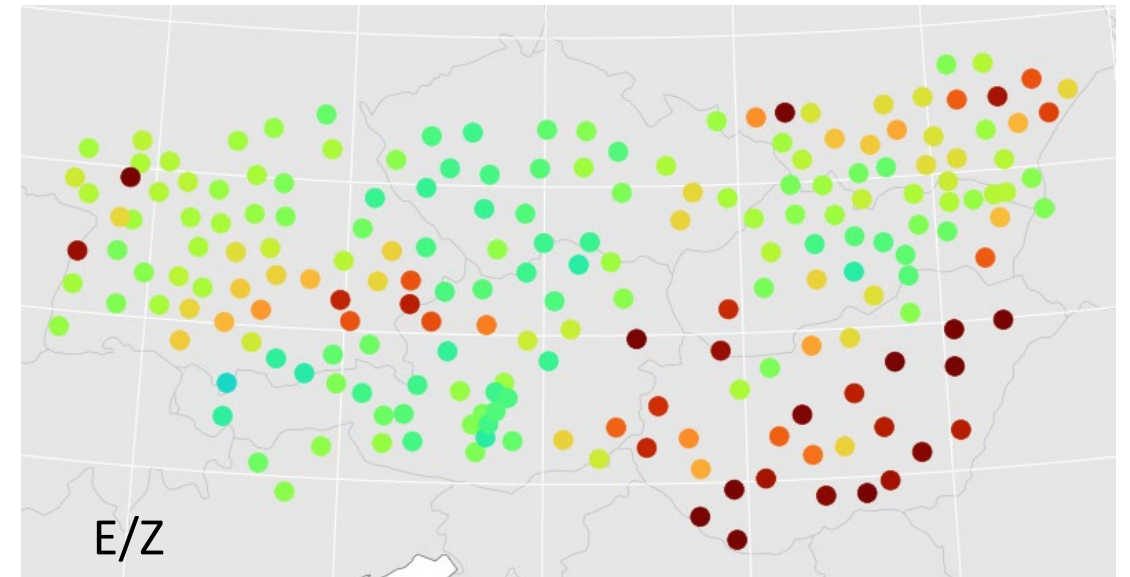
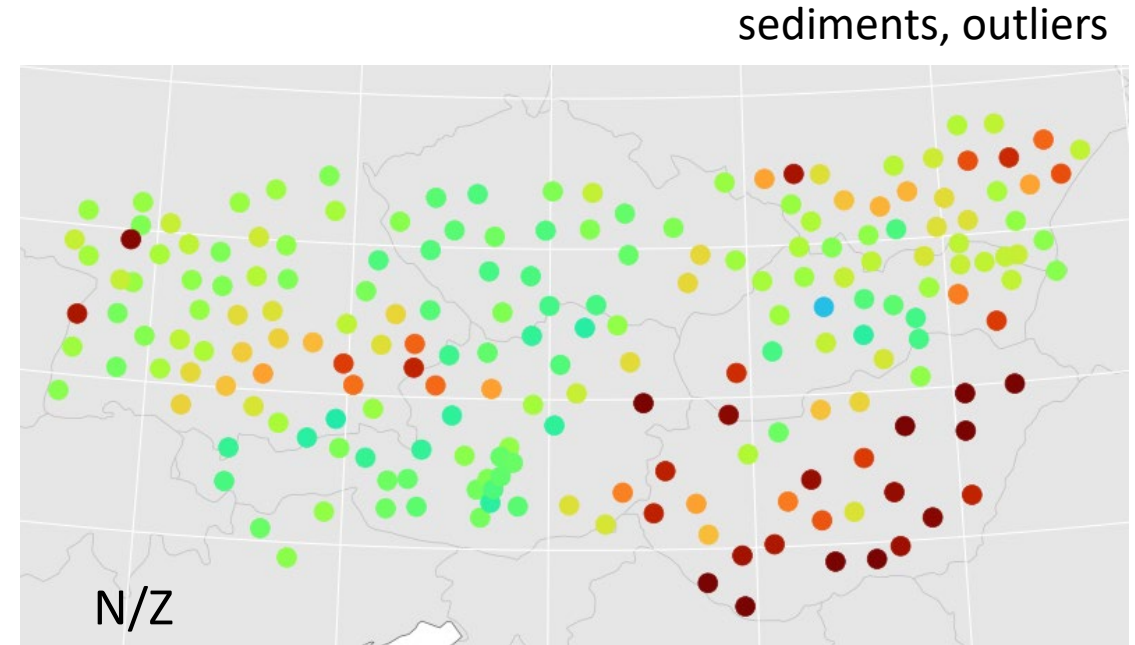
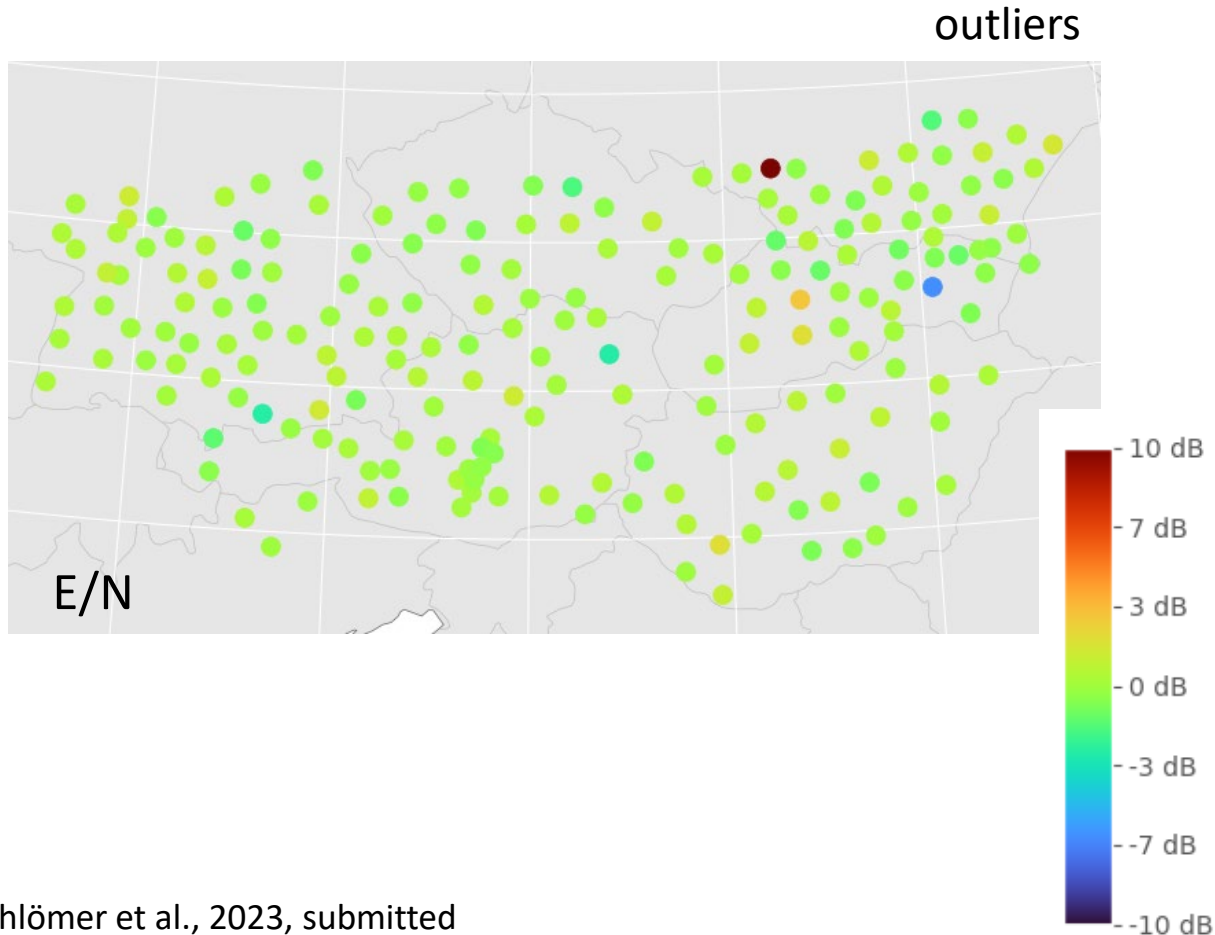


Control and calibration units



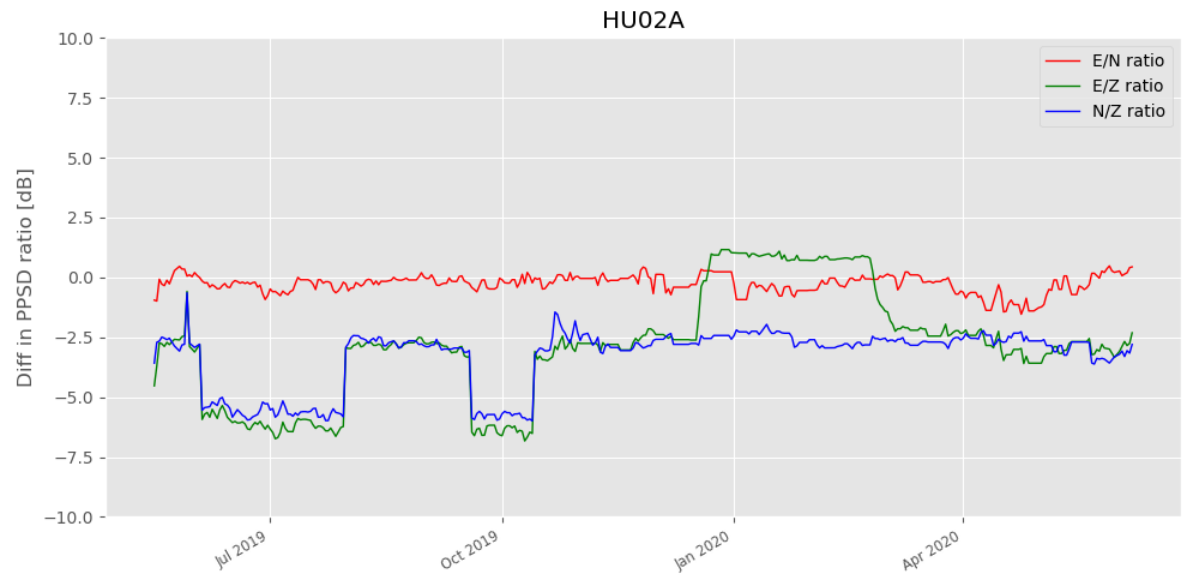
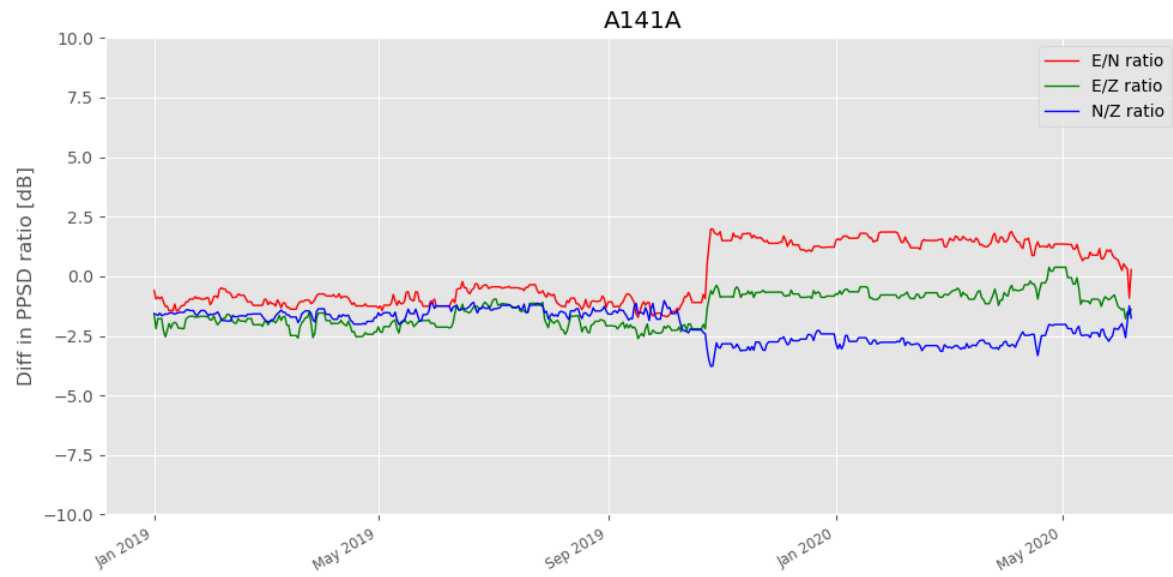
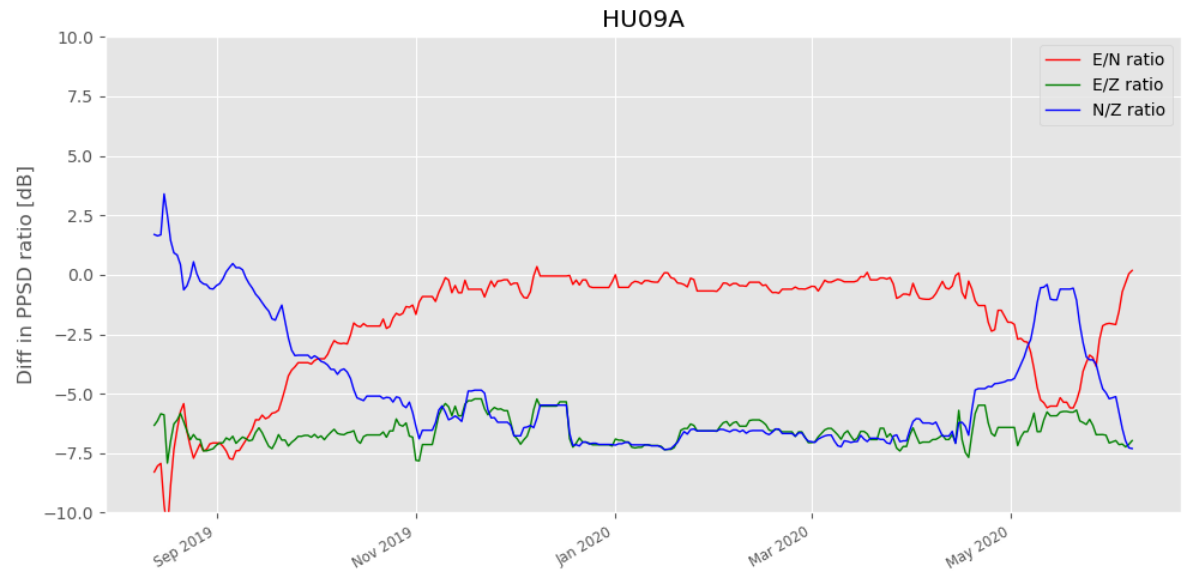
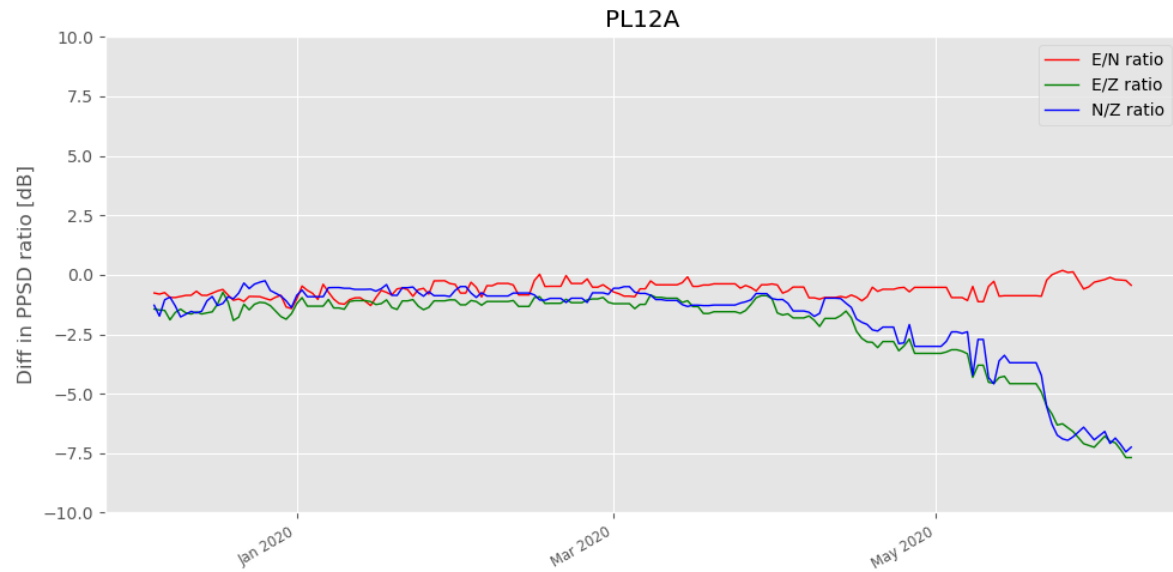
PACASE - amplitude gain issues

- average PPSD ratios for $T = 5$ s
- global trend subtracted



PACASE - amplitude gain issues

- temporal variations of PPSD ratios (for $T = 5s$)
- global trend subtracted



Sensor mis-orientation

How to measure:

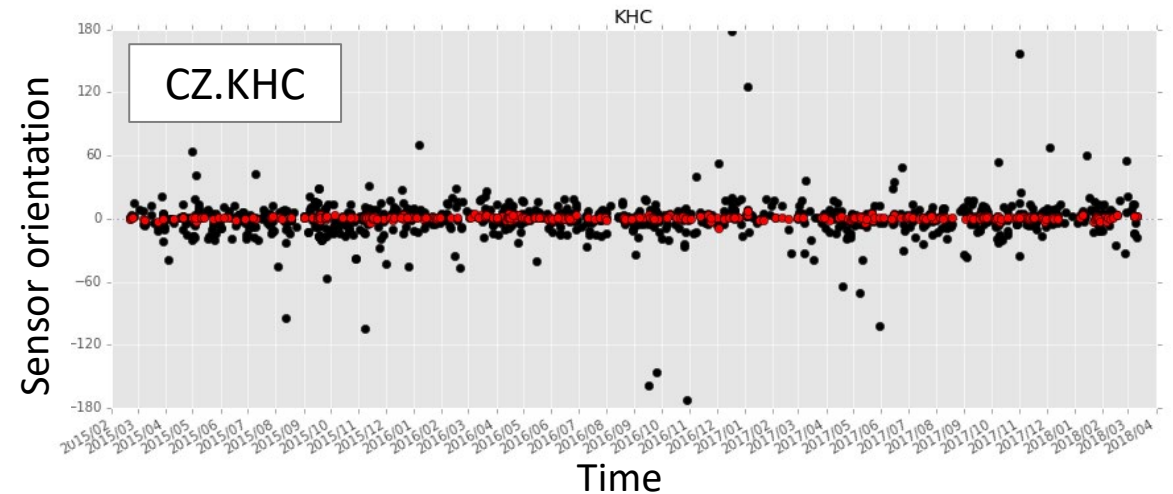
Gyrocompass: very accurate, not very frequent usage (if ever)

Polarization of Rayleigh waves (single station method): not so accurate, better detection of temporal changes (see black dots)

Improved Rayleigh wave method (multi station method): corrected for dynamic back-azimuths, best precision 1-2° (see red dots)

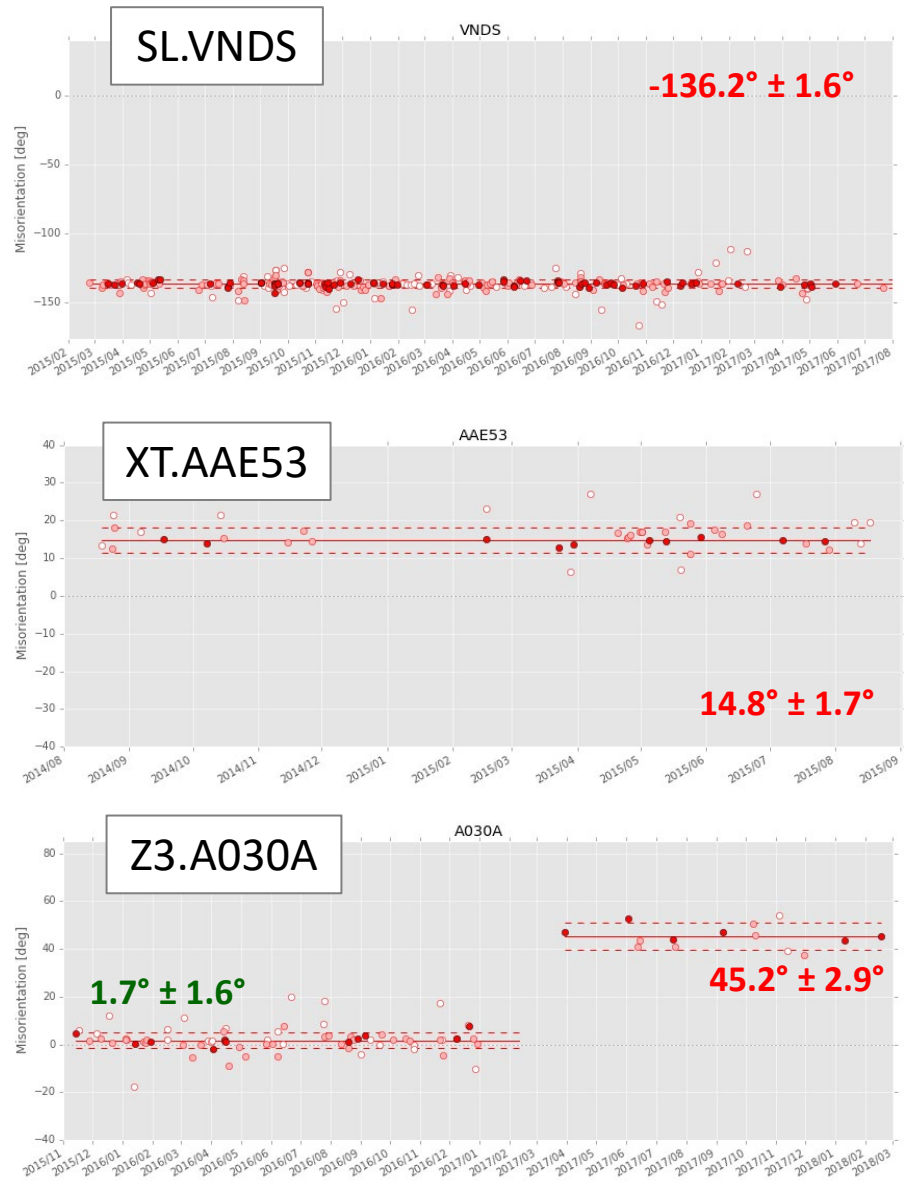
How to correct:

Change values in station metadata (easy!)



Comparison of sensor orientations re-measured by a gyrocompass and computed by the RW polarization method.	Gyrocompass		RW polarization	
	AAE27	8.1°	10.7°	± 3.3°
	AAE28	7.5°	8.5°	± 2.1°
	AAE29	13.2°	13.7°	± 1.3°
	AAE30	-3.6°	-4.0°	± 2.2° (+89°)
	AAE31	6.5°	7.7°	± 2.5°
	AAE32	-0.3°	3.2°	± 2.4°
	AAE33	8.1°	8.4°	± 1.4°
	AAE34	5.4°	6.0°	± 1.5°

Sensor mis-orientations



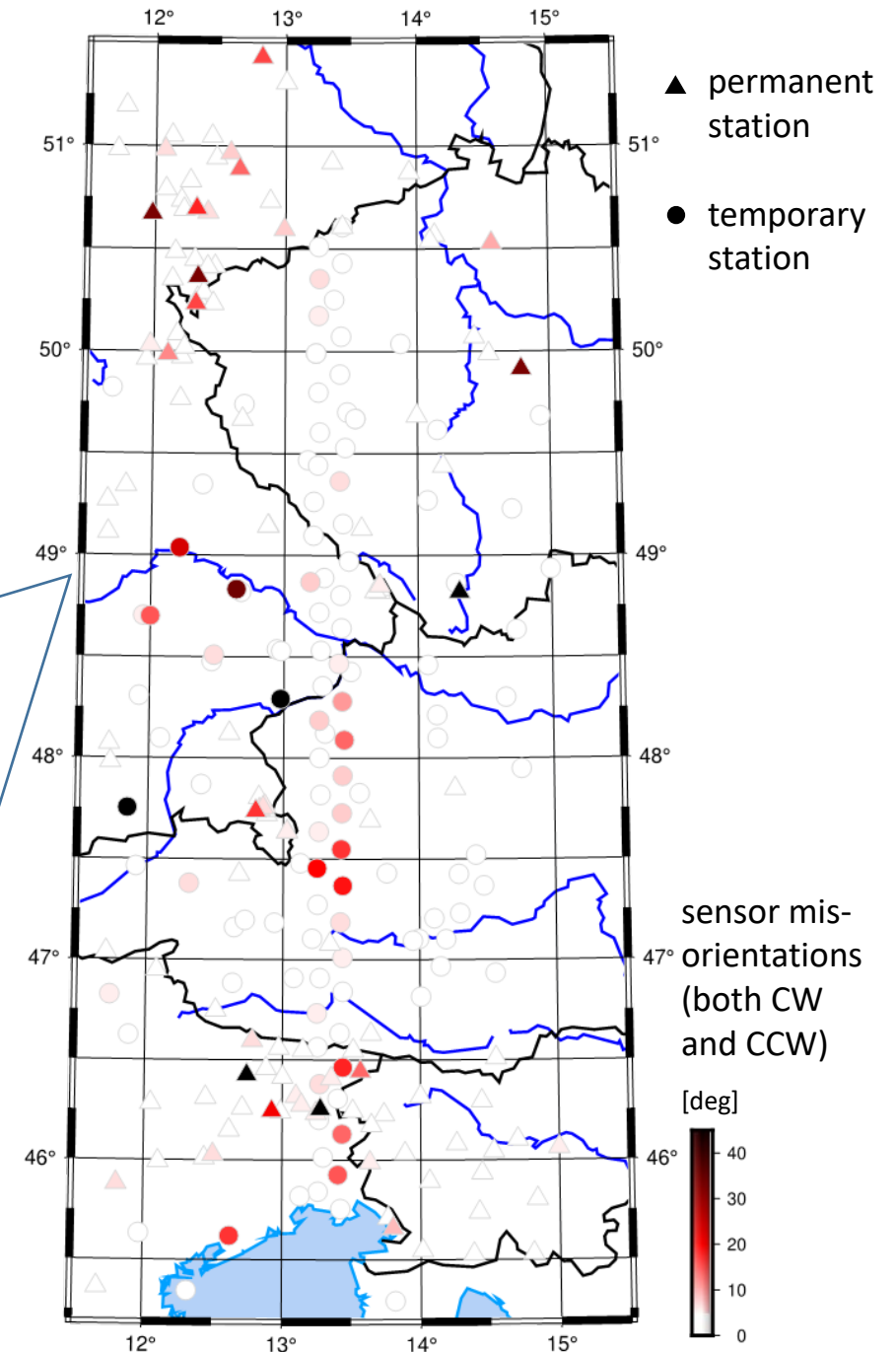
2014-2018: Rayleigh wave method (multi station)

2014-2023: SKS polarization method (single station)

Deviations of sensor orientations from the official orientations given in metadata (from EIDA, 2023-06-05).

Number of stations with sensor mis-orientations:

$\geq 5^\circ$	77/250	31 %
$\geq 10^\circ$	29/250	12 %
$\geq 20^\circ$	12/250	5 %
$\geq 30^\circ$	9/250	4 %
$\geq 45^\circ$	5/250	2 %
$\geq 90^\circ$	3/250	1 %



Conclusions

THINK QUALITY!
BEFORE
QUANTITY

- To guarantee high-quality data and metadata from MOBNET stations, we use a **double-check** for quality control: the hardware control in-situ and the ex-post software data checking
- Seismic network operators and data providers are responsible for quality data. Don't forget metadata!
- A non-negligible part of the data and metadata in EIDA is still defective or incorrect. Processes are underway to correct this condition
 - ❖ EIDA Issue Tracker - <https://github.com/EIDA/userfeedback/>
 - ❖ User Advisory Group (UAG) - reviews services and suggests improvements to ORFEUS services and products
 - ❖ AdriaArray - Working Group 3 'Data QC'

Thank you for your attention!

Acknowledgment

This research has been partly supported by the Grant Agency of the Czech Republic (grant no. 21-25710).

Several figures and calculations have been prepared with the use of the Generic Mapping Tools (Wessel and Smith, 1998) and ObsPy (Krischer et al., 2015).

We would like to thank the Seismic Network Teams of AlpArray, PACASE and AdriaArray, and the data providers of all permanent seismic networks in an AdriaArray region.